Technical Appendix:
An Empirical Analysis of Transfer for Dynamic Path Planning

Anonymous submission

A. Protocols by Stage

All continuous stages share the substrate: a 2-D point robot, maps with procedurally generated obstacles, a probabilistic roadmap (PRM) of $\sim$192 nodes with k=7 connectivity, and A* evaluated at a per-suite *binding budget* of node expansions calibrated so the algorithmic baseline is neither saturated nor hopeless. Unless noted, training labels are exact graph-Dijkstra costs on the PRM (static) or exact backward space–time Dijkstra values over (node, t) (dynamic). Headline continuous runs are local single-GPU (RTX 5090) validations with one primary training seed unless stated; the discrete program ran on a remote fleet.

A2. Exact Environment, Model, and Data Specifications

World generation. Continuous worlds are unit squares (side length 1.0; the large-room family uses 2.0); units are abstract. Start and goal are sampled with minimum separation 0.45 of the side length and inserted as roadmap nodes 0 and 1. Roadmaps contain exactly 192 sampled nodes with k=7 nearest-neighbor connectivity; edges are validated by collision sampling at resolution 0.010 of the side length, and worlds whose roadmap fails to connect start to goal are rejected and resampled.

Dynamic data collection. A candidate training world is *usable* iff (i) the world builds, (ii) its roadmap connects start to goal, and (iii) the space–time problem is solvable from the start at t=0 (finite backward space–time Dijkstra value). The heavy run draws seeds until 53 usable worlds are collected; each world contributes its reachable (v, t) states (all 192 nodes across $t_{\max}$+1 time steps, masked to reachable, averaging roughly 1.9×10^4 states per world), for 1,129,536 supervised scalar samples in total. All three training seeds use the same collection procedure with independent seeds.

B. Full Static and Dynamic Suite Tables

Tables 5 and 9 give the complete per-suite C7 static and C8 dynamic results summarized in the main paper; Tables 6–8 give the fixed-provider primary view and its replications.

Multi-seed replication detail (Table 8). Two full-pipeline retrains (collect + train, seeds 2001/2002) of the aware/blind field U-Net twins were run under a design frozen before training, then evaluated on the same frozen fresh cohort with the canonical binding budgets. Matched-ratio medians across seeds: maze 0.047–0.059, rooms 0.117–0.135, spiral 0.065–0.092 (seed-stable); crossing 0.273–0.434, dense maze 0.216–0.625, large rooms 0.250–0.548 (seed-variable). Preregistered readouts: R1 (success CI excludes zero in $\geq$5/6 suites) 5/6 and 6/6—pass; R2 (deltas within ±0.15 of canonical in $\geq$5 suites) 5/6 and 6/6—pass; R3 (no suite significantly aware-over-blind) fails as stated for seed 2001 (large rooms $+0.080$ $[+0.020, +0.160]$)—the same suite that is significantly *blind*-better under seed 1234 (-0.200 $[-0.320, -0.100]$). Across the 18 seed–suite twin contrasts: 2 significantly aware, 2 significantly blind, 14 null, with the significant cells in different suites for different seeds; the per-suite future-window effects flip sign across seeds and are consistent with training-seed noise. No retraining, reselection, or recalibration occurred in response to any readout.

Success as a function of budget (Figure 2). A single evaluation pass at 4$\times$ the binding budget on the same frozen cohort yields *exact* success-vs-budget curves for every lower budget: the space–time A* expansion order is budget-independent, and a map is solved at budget B iff its recorded solve expansions are $\leq B$, so the curves are derived by thresholding with no re-runs and no interpolation (design frozen before launch). As a determinism cross-check, thresholding at the binding budget reproduces every fresh-cohort success value exactly (all suites, all providers). The headline result is not an artifact of the calibrated operating point: Euclidean search needs 1.7–2.8$\times$ the binding budget to reach the fixed blind provider's binding-budget success (crossing 2.49$\times$, maze 2.29$\times$, dense maze 1.95$\times$, rooms 1.73$\times$, large rooms 1.94$\times$, spiral 2.80$\times$), and by 4$\times$ the binding budget every provider converges to the same feasibility ceiling (0.94–1.00; on dense maze the 0.94 ceiling binds even the oracle, reflecting maps unsolvable within the time horizon).

C. Transfer and Composition Details

Representative C9 HRM curves (ratio at success): maze-dense LoRA 0.650 (1.00) at K=1 to 0.730 (0.97) at K=16; maze-dense full FT 0.744 (0.76) to 0.571 (0.99) at K=16 and 0.552 at K=32; scratch 1.008 (0.37) to 0.808 (0.80); rooms-large full FT 1.128 (0.37) at K=1 to 0.500 (0.92) at K=8; bugtrap LoRA degrades from $\approx$0.696 (0.83) to 0.950 (0.48) at K=32—LoRA is usually but not universally base-preserving. C9H (matched compute): bounded-

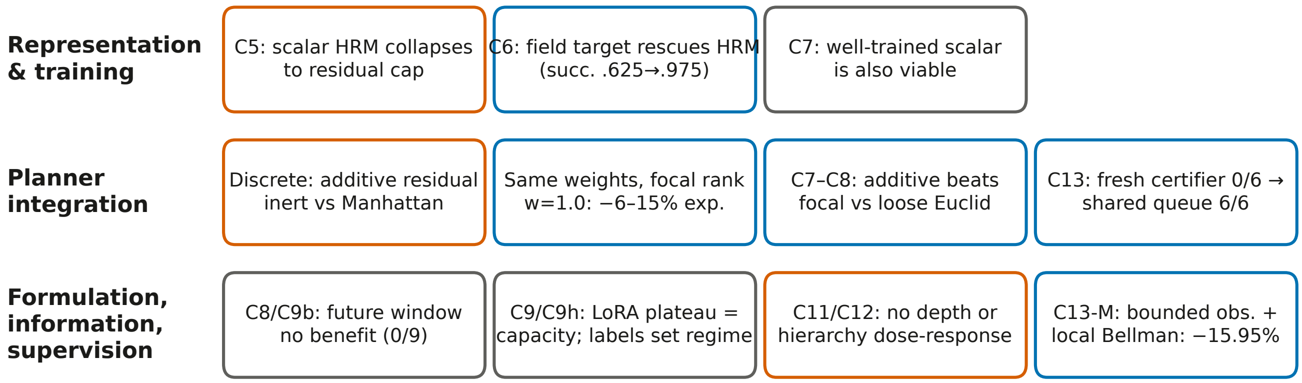

Figure 1: Program map: the three harness ingredients the study isolates—representation and training, planner integration, and formulation/information/supervision—with the stages that intervened at each and their verdicts (orange: failure or negative; blue: positive under the corrected harness; gray: mechanism or null result).

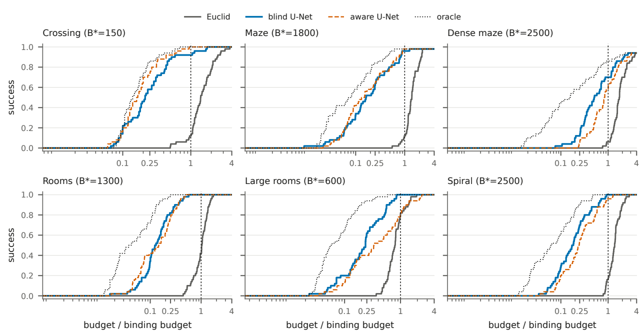

Figure 2: Success-vs-budget step curves per dynamic suite (fresh 50-map cohort), derived exactly from one $4\times$-binding evaluation pass. Budgets are normalized by the binding budget B^* (dashed vertical line). The learned providers reach high success at 0.05–0.3$\times$ B^*; Euclid requires 1.7–2.8$\times$ B^* to match the blind provider's binding-budget success.

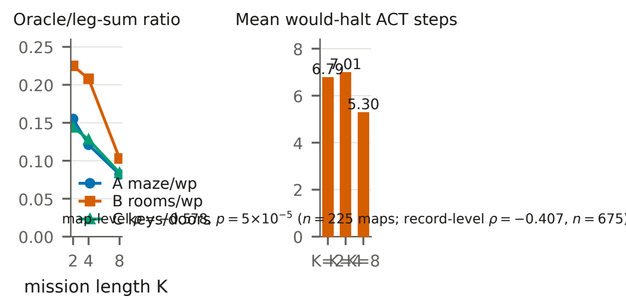

Figure 3: Compositional-mission hierarchy test (C11); K here is mission length. Left: the exact oracle's ratio over a leg-sum baseline falls as missions deepen (headroom exists). Right: learned HRM halting uses *fewer* steps at K=8; the annotation reports both analysis grains (map-level $\rho = -0.578$, $p = 5 \times 10^{-5}$, n=225 maps; record-level $\rho = -0.407$, n=675).

minus-unbounded LoRA median ratio delta 0.000 ± 0.008 across 27 cells (22/27 exactly zero; max $|\Delta|$ 0.028); rooms-large field U-Net full FT reaches 0.404 at 0.97 success from 0.982 zero-shot while conv-LoRA stays at 0.992–1.002. C9B (dynamics): maze-dense aware U-Net zero-shot 0.145 (0.60), LoRA K=16 0.059 (0.97), full FT K=1 0.112 (0.70); aware-minus-blind success deltas at full-FT K=16 are 0.00 or -0.05 in all nine cells. C10: RBF-merge minus zero-shot expansion-ratio deltas by target/backbone: -0.022, -0.012, -0.024, $+0.082$, $+0.012$, $+0.116$; weight-space vs. prediction-space deltas lie within $[-0.012, +0.007]$; own-axis RBF mass 0.986–0.998.

D. C11 Gates, Collapse Forensics, and Scale

Collapse forensics. Five of 33 recurrent/ACT arm-cells show constant-prediction or partial seed collapse (HRM-trace A8/C8, ON-LSTM B0, HRM-v2 B0, two ON-LSTM A8 seeds), with a reproducible loss signature at the normalization cap. The largest recurrent deficits (HRM-trace ratios 1.526 at A8, 1.600 at C8) are collapse measurements, not expressiveness estimates. Removing K=0 padding *worsens* recurrent predictions: models used pad steps as extra recurrent compute.

Scaled addendum. A 12-run `unet_film_big` vs. `hrm_trace_big` comparison (complete; 2,700 eval rows) preserves the ordering: completion 0.804 vs. 0.736 at K=2 and 0.413 vs. 0.307 at K=8; mean state MAE 0.395 vs. 0.447 (K=2) and 0.906 vs. 1.855 (K=8). At K=8 the scaled HRM exactly matches the leg-sum baseline's completion and mean expansions (0.307; 701.34)—persistent collapse, not recovered hierarchy.

E. C12 Detail

C12-A (memory headroom, then learned pilot): constructed decision-relevant aliasing at 71.3% of paired decisions; privileged mode/history diagnostic +0.467 completion and -65.1% collision-adjusted regret (map-bootstrap 95% CI 63.0–66.9%); oracle completion 0.975. The one-seed matched temporal-hierarchy pilot fails G1 (forecast), G2 (planning), and G3 (persistent carry); overall *strong negative*. C12-B (tied refiner): cycle-1→8 normalized burden improves monotonically in every cell (A/$K2$ 0.859→0.815; A/$K8$ 0.624→0.576; C/$K2$ 0.867→0.777; C/$K8$ 0.576→0.517; all cycle-1/8 map-bootstrap intervals separate), but the $K8-K2$ gain is $+0.0038$ (CI $[-0.0199, 0.0281]$) on A and -0.0305 ($[-0.0726, 0.0096]$) on C—no dose–response. Tied beats shallow and untied on C/$K8$ (deltas 0.0177/0.0076, BH $q=6.7\times10^{-5}$) but loses to untied on A/$K8$ ($q=0.0168$). Bellman residual *worsens* over cycles in every cell while burden improves; solved cycle-8 paths average 1.4–2.2% above oracle cost. $K=16$ was dropped before training by a construction gate (only 2/300 and 1/300 valid missions).

F. C13: Mechanism Ladder and Frozen Confirmation

Claim supported. A provider restricted to bounded local observations (radius-0.20 subgraph on a known roadmap; behavior-derived labels; no shortest-path supervision) can, with one inference-time local Bellman backup, beat every fixed complete-map C7 provider in pooled expansions at matched empirical path quality on a frozen, disjoint cohort.

Ladder. Each rung changes one factor; deltas are paired expansions versus the matched comparator (rungs C–G versus matched focal search at $w=1.10$, including oracle providers on six development maps; rungs I–M versus the complete-map field HRM, live on all six suites).

Frozen confirmation, per suite. 144 maps (24 per family; seeds disjoint from every prior cohort; feature caches hash-verified against live regeneration; model, radius 0.20, $\alpha=1.50$, integration, and analysis frozen before generation).

Scale calibration. On the disjoint 48-map calibration block, $\alpha \in \{1.00, 1.25, 1.50, 2.00\}$ gives deltas -3.79 $[-9.40, +1.73]$, -9.77 $[-15.38, -4.42]$, -13.04 $[-18.67, -7.65]$, and -16.94 $[-22.56, -11.54]$, with 4/4/6/6 negative suite means and mean/max cost ratios rising from 1.022/1.241 to 1.052/1.366; $\alpha=1.50$ was frozen as the largest scale whose suite breadth was complete before the absolute-ceiling verdict, and every arm formally fails the preregistered absolute 1.10 maximum-cost ceiling—the confirmation gate is therefore comparator-relative (mean within $+0.005$, max within $+0.02$ of field HRM), which passes with margins to spare (1.0235/1.1624 vs. 1.0311/1.3346).

Operating points and timing. The direct $\alpha=1.50$ arm is the matched-quality expansion result and carries no formal bound. The separate certified operating point runs the learned rank inside a $w=1.10$ focal band with a shared-queue consistent-bound incumbent proof and passes all 144 bound and certificate checks. Per-map wall time decomposes as feature construction 5.138 s (prototype Python builder), model inference 0.0008 s, local backup 0.0097 s, and search 0.0002 s, versus 0.371 s complete-map field-HRM inference: the result is search effort and path quality, not latency.

Architecture substitutions under the identical method (C13-N/O/P) are summarized in the main paper; their raw artifacts (development grids, verdicts, integrity manifests) ship in the artifact package.

G. Discrete Program Detail

The discrete program is substantially larger than the canonical subset reported here: fifteen experiment families across eleven surveyed cloud volumes (13,671 parsed result files) and roughly thirty experiment scripts, including a ten-run early scaling lineage of obstacle-trajectory forecasters, diffusion path-generator baselines, a superseded harder-calibration predecessor of the structured protocol below, partial precursors of the controlled transfer run, an empty-model baseline-rerun control (static A^* re-verified through the learned-model code path), and a budget-invariance analysis establishing that raising search budgets adds work without rescuing success on the hard families. The paragraphs below report the completed canonical results; the full inventory ships with the released artifacts.

Forecasting. Matched 100-episode protocol: HRM-3M/10M 71/100 vs. LSTM 67–69/100 (no uncertainty reported; two-episode matched gap). Structured Preset M+ (four 100-episode suites): ON-LSTM-10M 0.3875 mean success vs. HRM-10M 0.2650 and HRM-3M 0.200; ON-LSTM-3M also has the lowest rollout MSE (0.1257/1.1269 one-/five-step vs. 0.3973/2.6438 for HRM-3M).

Additive transfer (clean v3, 22 suites $\times$ 3 budgets). Manhattan 0.590 success / 126,707 mean expansions; HRM full-FT 0.585 / 128,013; ON-LSTM full-FT 0.564 / 127,228; HRM LoRA 0.554 / 140,852; ON-LSTM LoRA 0.514 / 145,282. Every learned arm $\leq$ matched baseline.

Pooled multitask. HRM pooled base 0.612 success (+2.11pp) / 122,184 expansions vs. Manhattan 0.591 / 126,630—the strongest completed discrete additive positive; only 1 of several specialists beats its matched four-suite baseline (+2.0pp; others -0.42 to -20.25pp).

Focal ranking (local pilot, 3–8 seeds). Magnitude is scale-flat despite near-perfect ordering (correlation 0.987–0.994; predicted/true mean 0.73/0.42/0.27/0.19 at 64/128/192/256). At $w=1.0$: HRM A128-static ratio 0.85 with success 0.62→0.75; A192 0.94 (0.75→0.75); A128-dynamic 0.93 (0.62→0.62); ON-LSTM A128/A192 0.85 (0.75→0.75). At $w=1.05$, A128-dynamic improves further but A192 success regresses 0.75→0.62; $w=1.0$ is the evidence-safe operating point. Experts exactly match pooled bases in tested cells.

H. Validity Forensics

The frozen-halt artifact. A 27.27M-parameter direct maze solver reported 96.64% token accuracy, 25.40% exact accuracy, and "74.60% halt accuracy" with 16 average adaptive-computation steps. Audit: $74.60 = 100 - 25.40$ exactly—the halt logit was frozen negative (a constant-false classifier) because the training loop omitted the halt loss and back-propagated only the final segment. After repairing the loss path (verified live: halt loss 0.14→0.06, token accuracy 90–95%), exact accuracy remained 0.000 at the chosen stop point; the repaired run demonstrates mechanism liveness, not task convergence. Bit-exact forward parity with the reference implementation was separately established. We recommend the $100-x$ pattern as a standard red flag for adaptive-computation metrics.

The cap-collapse signature. C5's scalar HRM collapse sits at a loss value pinned by the residual cap (2.612 in that setup, with constant predictions at the cap). The same class of signature reappears in C11: collapsed cells share an *identical* final loss (1.1397, exactly the constant-cap loss under that normalization), enabling automatic collapse detection instead of misreading collapsed cells as capacity evidence.

Pad steps as compute. In C11, removing $K=0$ padding tokens *degraded* recurrent predictions—models had co-opted padding as additional recurrent iterations. Sequence-model comparisons that alter padding alter compute.

I. Statistics

Success is compared with paired McNemar tests, BH-corrected within stage families. Expansion effort is compared only on matched solved maps via median ratios with bootstrap percentile CIs (10k–20k resamples); from C12 onward, bootstraps are map-clustered by design. C9–C11 originally pooled repeated evaluations of fixed test maps across adaptation/model seeds; Section J reports the map-clustered reanalysis of every quantity the main paper quotes from those stages; the fixed-provider dynamic reanalysis of Table 6 uses the same map-level grain. Matched-solved ratios condition on joint success and can favor arms that solve only easier maps; we therefore always report success and ratios separately and never combine them into a single score. Negative results are failures to observe an effect under the tested protocol, not equivalence claims; no equivalence tests were run.

J. Map-Clustered Reanalysis

Units are test maps; adaptation/model seeds are averaged within maps before inference; success deltas use 10,000-resample map bootstraps; expansion ratios are medians of per-map ratios over matched maps at the binding budget in additive mode; the C11 halting test aggregates model seeds within maps and permutes mission length within config (maps are independent across K cells; 20,000 stratified permutations). Estimator conventions differ slightly from the published record-level curves, so point values differ modestly; the question is verdict-level agreement. A methodological caution surfaced during this work: a first pass that accidentally pooled additive and focal evaluation modes attenuated the apparent C9 transfer effects by roughly half—mode/aggregation mixing alone can manufacture or destroy an effect of the size under study.

C11 halting. Record level reproduces the published value ($\rho=-0.4066$, n=675); map level strengthens it ($\rho=-0.5778$, n=225, stratified permutation p=5×10^{-5}). Map-level means 6.78/7.01/5.30 at $K=2/4/8$.

C11 vs-MLP ratios. Map-clustered ratio-of-means CIs exclude parity at $K\in\{0,2\}$ in config A for both U-Net (0.734/0.846) and GNN (0.822/0.923), and in config B at K=0 (0.904/0.818) plus B/K=2 for the GNN (0.805); config C never separates; every $K\in\{4,8\}$ cell includes parity.

C9H bound-vs-capacity. 27 cells: median bounded-minus-unbounded map-level ratio delta $+0.0000$; 15/27 exactly zero; sd 0.0077; max $|\Delta|$ 0.0370—the capacity attribution stands.

C9B aware−blind (full FT, K=16). Eight of nine point deltas are zero or negative; the single positive ($+0.017$ $[0.000, +0.050]$) does not exclude zero; no cell shows a significant aware advantage.

C10 composition. Map-paired deltas vs. zero-shot: four cells ≈0 with CIs crossing zero; the two ON-LSTM rooms cells are significantly *worse* ($+0.089$ $[+0.061, +0.121]$ and $+0.098$ $[+0.076, +0.123]$), matching the published point deltas and upgrading the null to "never better, sometimes significantly worse."

No preregistered verdict flips under any of these recomputations.

J2. Fixed-Model Selection Audit Trail

The primary dynamic claim rests on one fixed learned heuristic; its provenance is as follows. (1) *Candidates:* the original dynamic run trained aware and blind twins of scalar HRM, scalar ON-LSTM, and field U-Net models. (2) *Selection:* the field U-Net blind twin was designated the single primary model for every suite during the fixed-model reanalysis design; the original 20-map evaluation results were known at that time, so the accurate statement is that the model, integration mode (additive), and per-suite binding budgets were **fixed before the 50-map confirmation cohort was generated**, not that selection predates all target outcomes. (3) *Freeze:* the confirmation cohort's generation seed (999,999) was chosen in advance; the evaluation-seed formula's within-suite offset bound guarantees map-disjointness from every earlier cohort, and world fingerprints (start, goal, obstacle coordinates) were checked. (4) *Integrity:* the fixed checkpoint's SHA-256 is `b8378950545f8abdbf06d59568b5d8ab6069884c1c038d23a41a14b6dc17fb6f`; the binding-budget calibration file is copied verbatim into every evaluation directory. (5) *Seeds:* the two additional training pipelines change only the training seed (2001, 2002); collection, architecture, integration, budgets, and the evaluation cohort are identical.

K. Reproducibility and Artifact Inventory

From C7 onward every stage freezes a written design (gates, budgets, analysis grain) before execution; artifacts (checkpoints, feature caches, cohort manifests, raw rows, verdicts, reports) are hash-bound in integrity manifests; confirmation cohorts are generated only after freezing model, integration, and evaluation code; failed development gates block confirmation rather than triggering retuning. Duplicate-evaluation determinism is enforced where preregistered (byte-identical traces; identical decision rows). The final-stage pilot on persistent planning-time state ran under a fully test-verified harness (158 focused-plus-neighboring tests; hardware preflight). Its first staged execution ended at a preregistered execution-integrity hard stop—the harness's independent reanalysis disagreed with the primary comparison computation, voiding all mechanism fields. After repair of the named defect only, a fresh-fingerprint rerun passed every integrity primitive, including independent reconstruction of the comparison evidence from promoted raw artifacts, and returned the valid negative reported in the main paper: persistent carry loses to the same-checkpoint reset twin on map-macro frontier ranking (MRR delta -0.029, 95% CI $[-0.060, -0.003]$), and no self-bootstrap or confirmation was run. We note the initial stop as the integrity machinery functioning as designed.

The anonymized artifact archive submitted with this paper contains: (i) the analysis scripts that regenerate every main-paper figure and quoted map-level statistic from raw rows (the fixed-provider dynamic reanalysis, the map-clustered reanalysis, and all figure generators); (ii) the raw paired evaluation rows behind every quoted quantity (C7/C8 evaluation tables, including the fresh-cohort and three-seed replication rows, C9/C9H/C9B/C10 evaluation tables, C11 evaluation and halting tables, and C13 development and confirmation rows); (iii) map/cohort manifests, seed lists, frozen configurations, and the integrity manifests for the frozen confirmations; (iv) the preregistration design documents from C7 onward; and (v) a top-level README with reproduction commands and expected outputs. Checkpoints are included where size permits; otherwise exact training commands and configurations are provided.

L. C14: Label-Count × World-Diversity Factorial

A preregistered factorial (design and pre-execution amendment in the artifact package) tests whether the LoRA/full-fine-tuning contrast is governed by the number of supervised search states N, independent of domain and map count. Factors: $N \in \{256, 1024, 4096, 16384, 65536\}$ exact-count state samples × two diversity levels at fixed N (concentrated: the minimal world count $w_{\min}(N)$; distributed: $8 \times w_{\min}(N)$ worlds) × two domains (static C9 substrate and dynamic C9B substrate, maze-dense target, frozen sources) × {unbounded rank-8 LoRA, full fine-tuning, scratch} × 3 seeds — 180 adaptations, every cell at an identical 2,560 optimizer steps, evaluated on the frozen C9/C9B test cohorts at binding budgets. The measured $w_{\min}$ spans quantify the domains' supervision densities: reaching N=65,536 takes 356 static worlds but 4 dynamic worlds (∼185 vs. ∼18,900 usable states per world).

Preregistered verdict: H-C14 rejected as stated. On matched-solved expansion ratios the hypothesized crossover point is degenerate: full fine-tuning is at or below LoRA at *every* tested N in all 12 (domain × diversity × seed) cells, and the preregistered regression's full-FT × $\log N$ interaction is null (-0.0025 $[-0.0087, +0.0028]$). State count improves ratios ($\log_2 N$ coefficient -0.0140 $[-0.0190, -0.0081]$) but identically for both methods.

The preregistered diversity readout localizes the real effect. The low-supervision protection LoRA provides appears in *success*, and it is governed by world diversity at fixed N, not by state count. Concentrated low-world cells collapse full fine-tuning and scratch in both domains (success vs. the frozen zero-shot source: -0.300 $[-0.550, -0.050]$ in all three seeds at dynamic concentrated N=256 and N=1,024; -0.167 to -0.300 at static concentrated N=256), while the distributed cells at the same N rescue them (dynamic $+0.150$ to $+0.200$; static ≈ 0.000; paired dist−conc full-FT deltas $+0.30$ to $+0.48$ dynamic, $+0.233$ static at N=256). LoRA never collapses in any of the 60 cells. The dynamic concentrated arm recovers exactly where $w_{\min}$ forces multiple worlds (N=65,536, 4 worlds: $+0.100$ in all seeds), tying the recovery to world count at fixed steps. This refines, without contradicting, the main text's supervision-density account: in the K-indexed adaptation protocols map count and state count move together, and C14 separates them — the protective variable is diversity (or adapter-restricted capacity), not raw state count.

Limitations. Euclid solves 1/20 dense dynamic test maps at the binding budget, so dynamic matched-ratio cells have n=1 and carry no map-level uncertainty; dynamic inference rests on the 20-map paired success deltas. One target family per domain; three adaptation seeds; adaptations ran on cloud L4 hardware from locally collected, hash-recorded datasets.

Stage	Protocol essentials
C1–C4	9 easy suites, n=80/40 maps, budgets 100–200; pooled bases, TaskLoRA experts, RBF mixtures.
C5	calibrated hard maze/dense/rooms; 160 train / $\sim$80 eval maps; budgets 128–168; scalar residual target with cap.
C6	goal-conditioned raster residual field (8 input channels), sampled at PRM nodes; 96/40 maps; budgets 128–168.
C7	6 suites (3 held-out families); 96/24 maps per suite; binding budgets 140/24/56; scalar+field $\times$ additive+focal matrix; exact-Dijkstra oracle.
C8	deterministic moving circular obstacles; space–time PRM; aware (W=8 future window) vs. blind (W=0) twins; 53 train / 20 eval maps/suite; 1,129,536 scalar samples.
C9/C9H	adapt C7 pooled bases to 3 held-out families; $K \in \{0..32\}$ labeled maps; LoRA(r8, bounded/unbounded), full FT, scratch; C9H matches compute (10 epochs, LR 2×10^{-4}); 30 fixed test maps/target.
C9B	same protocol on C8 dynamic sources (aware+blind), 486 adapters, 3 targets, $K \in \{1,4,16\}$.
C10	16 rank-8 experts on 2 family axes; zero-label transfer to 3 interior targets; RBF/nearest/uniform weight-merge and prediction-mix composition.
C11	compositional missions (waypoint sequences; key/door), configs A/B/C, mission length $K \in \{0,2,4,8\}$; 40 train / 25 test maps per cell; 6 learned arms $\times$ 3 seeds (198 checkpoints; 54,450 eval rows); exact mission oracle and leg-sum baseline; forced-compute and would-halt probes.
C12	(A) hidden slow/fast regime dynamics, memory-headroom gate, matched temporal-hierarchy pilot; (B) tied vs. untied product-graph refiner, cycles 1–8, configs A/C $\times$ $K \in \{2,8\}$, 3 seeds, 48 checkpoints, two 3,200-row test tables.
C13	current-state heuristic under bounded observation (radius 0.20 local subgraph); behavior-derived fresh-start rollout labels (median of 3); staged falsification ladder (A–L) then frozen 144-map confirmation (M); architecture substitutions (N/O); persistent-state pilot (P).
Discrete	20$\times$20 to 256$\times$256 grids; Manhattan-baseline A*; forecasting (dynamic obstacles), additive residual transfer (22 suites $\times$ 3 budgets $\times$ 100 episodes), pooled multi-task bases, learned focal ranking at $w \in \{1.0, 1.05, 1.1\}$.

Table 1: Protocol essentials by stage.

Family	Generator rule (lengths as fractions of side length)
Maze / dense maze	Corridor maze of axis-aligned walls; the dense variant narrows gap width to 0.115. Held out: dense.
Rooms / large rooms	2$\times$2 rooms from one vertical + one horizontal partition wall, one doorway per wall, doorways biased toward opposite edges; large variant doubles side length. Held out: large.
Spiral	Four serpentine vertical walls spanning $x \in [0.22, 0.78]$, gap side alternating per wall. Trained.
Bugtrap	Solid vertical back wall, two horizontal arms, and a mouth lip closing the center of the opening (U-pocket). Held out.
Crossing (dyn.)	Open arena; dynamic timing control.

Table 2: Static geometry per family. Base-anchor families (open/clutter/narrow variants) used by the early stages additionally place 3–68 circular obstacles with radii 0.022–0.115; full ranges ship with the generator in the artifact package.

Suite	Patr.	Radius	Span	Period	v_{ag}	$t_{\max}$
Maze	3	0.075	0.38	0.14	0.060	110
Rooms	3	0.075	0.40	0.14	0.060	110
Spiral	3	0.075	0.38	0.14	0.060	110
Dense maze	4	0.062	0.42	0.17	0.060	140
Crossing	4	0.055	0.50	0.30	0.060	110
Large rooms	5	0.075	0.36	0.14	0.120	130

Table 3: Dynamic-suite parameters. Each patroller is a circle sweeping a line segment with triangle-wave motion; radius and span are fractions of the side length, period is a fraction of the horizon $t_{\max}\, dt$ (dt=1 s), and each patroller crosses its span twice per period. Sweep centers carry lateral jitter ≤ 0.02 (0.10 on crossing). The planner and collision checker always use these exact trajectories; only the *aware* heuristic input sees the W=8 future-occupancy channels.

Model	Configuration	Input	Parameters
Scalar HRM	hid. 192, 2 layers, 4 heads, k=2	62 tok. $\times$ 16	2,158,529
Scalar ON-LSTM	hid. 256, 2 layers, chunk 8	62 tok. $\times$ 16	1,252,481
Field U-Net	FiLM-conditioned, 8 input channels	64$\times$64 raster	3,702,370
Field HRM	8 input channels	64$\times$64 raster	4,016,642
Field ON-LSTM	8 input channels	64$\times$64 raster	1,448,578

Table 4: Instantiated architectures with parameter counts computed from the released constructors. Scalar token sequences: one summary token, 12 nearest-obstacle tokens, 48 ray tokens, one goal token. Dynamic variants append W future-occupancy channels (W=8 aware; W=0 blind). LoRA inserts rank-8 adapters on the HRM (rank-24 on the ON-LSTM, matching its wider hidden layer); the quoted crossover cells are HRM (rank 8). Optimizer for all models: AdamW, learning rate 2×10^{-4} (1.5×10^{-4} for adapters), weight decay 10^{-4}, batch 256, gradient clip 1.0, smooth-L1 loss on the capped residual (cap 4.0), 10 base epochs (8 adapter epochs) unless a stage states otherwise.

Suite	Euclid succ.	Field-HRM succ.	Ratio [95% CI]
Maze	0.583	1.000	0.521 [0.450, 0.627]
Maze-dense	0.250	0.958	0.804 [0.707, 0.829]
Rooms	0.375	0.958	0.829 [0.775, 0.885]
Spiral	0.250	0.917	0.850 [0.742, 0.919]
Bugtrap	0.458	0.750	0.714 [0.533, 0.800]
Rooms-large	0.417	0.750	0.839 [0.646, 1.222]

Table 5: C7 static suites at binding budgets: success and matched-solved median expansion ratio (field HRM vs. Euclid). Success gains are BH-significant in 4/6 suites (maze, dense, rooms, spiral); bugtrap and rooms-large reach corrected significance only for other learned arms. Scalar-HRM ratios span 0.427–0.822. Matched n ranges 6–14.

Suite	Euclid succ.	Blind U-Net succ.	Ratio [95% CI]	n
Crossing	0.30	0.95	0.191 [0.166, 0.878]	6
Maze	0.30	1.00	0.093 [0.057, 0.137]	6
Dense maze	0.05	0.75	0.246 (descriptive)	1
Rooms	0.30	1.00	0.099 [0.050, 0.214]	6
Large rooms	0.75	0.95	0.228 [0.185, 0.418]	15
Spiral	0.20	0.95	0.087 [0.061, 0.287]	4

Table 6: C8 fixed-provider result on the *original* 20-map cohort (field U-Net blind; no future-motion input; additive mode; binding budgets; map-level bootstraps). Success deltas are $+0.20$ to $+0.75$ with all six paired CIs excluding zero.

Suite	Euclid succ.	Blind U-Net succ.	Ratio [95% CI]	n
Crossing	0.12	0.92	0.273 [0.109, 0.516]	6
Maze	0.12	0.96	0.047 [0.021, 0.153]	6
Dense maze	0.06	0.70	0.216 [0.100, 0.281]	3
Rooms	0.42	1.00	0.121 [0.106, 0.172]	21
Large rooms	0.82	1.00	0.250 [0.188, 0.306]	41
Spiral	0.16	1.00	0.092 [0.065, 0.146]	8

Table 7: C8 fixed-provider *replication* on the fresh 50-map-per-suite cohort (generation seed 999999, frozen before any result was observed; map-disjoint from the original cohort by the evaluation seed formula's within-suite offset bound). Success deltas are $+0.18$ to $+0.84$ with all six paired CIs excluding zero. Map-paired aware$-$blind success deltas on this cohort: crossing $+0.080$ $[+0.020, +0.160]$, maze $+0.020$ $[0.000, +0.060]$, dense maze -0.120 $[-0.220, -0.040]$, rooms 0.000, large rooms -0.200 $[-0.320, -0.100]$, spiral -0.040 $[-0.100, 0.000]$: one suite significantly favors the aware twin, two the blind twin, three neither.

Suite	Seed 1234	Seed 2001	Seed 2002
Crossing	$+0.80$ [0.68, 0.90]	$+0.86$ [0.74, 0.96]	$+0.88$ [0.78, 0.96]
Maze	$+0.84$ [0.74, 0.94]	$+0.86$ [0.76, 0.96]	$+0.84$ [0.74, 0.94]
Dense maze	$+0.64$ [0.50, 0.78]	$+0.46$ [0.32, 0.60]	$+0.56$ [0.42, 0.70]
Rooms	$+0.58$ [0.44, 0.72]	$+0.58$ [0.44, 0.72]	$+0.58$ [0.44, 0.72]
Large rooms	$+0.18$ [0.08, 0.30]	$+0.10$ [-0.02, 0.22]	$+0.16$ [0.06, 0.26]
Spiral	$+0.84$ [0.74, 0.94]	$+0.84$ [0.74, 0.94]	$+0.84$ [0.74, 0.94]

Table 8: C8 preregistered multi-seed replication: paired success deltas (blind U-Net $-$ Euclid, map-level 95% CIs) for three independent training seeds on the same frozen fresh cohort. 17 of 18 CIs exclude zero; the single crossing interval is seed 2001's near-ceiling large-rooms cell (Euclid 0.82).

Suite (arm)	Succ. E$\rightarrow$L	Ratio [95% CI]	Matched n
Crossing (U-Net)	0.30$\rightarrow$1.00	0.258 [0.132, 0.769]	6
Maze (U-Net)	0.30$\rightarrow$1.00	0.064 [0.046, 0.088]	6
Maze-dense (U-Net)	0.05$\rightarrow$0.75	0.278 (no CI)	1
Rooms (U-Net)	0.30$\rightarrow$1.00	0.096 [0.038, 0.191]	6
Rooms-large (U-Net)	0.75$\rightarrow$0.90	0.380 [0.132, 0.591]	14
Spiral (field HRM)	0.20$\rightarrow$0.90	0.046 [0.038, 0.073]	4

Table 9: C8 *secondary* view: per-suite strongest arms (explicitly post-hoc selection). Corrected success evidence holds on five suites; rooms-large is supported mainly by expansion evidence. Dense-maze has matched n=1 and supports no expansion inference. Blind arms are the per-suite best in five of six suites.

Gate	Verdict
K=0 continuity	FAIL, diagnosed as false premise: C7 had no MLP arm and did not give every architecture the C11 input; real shallow separation plus collapsed recurrent cells violate the all-tie assumption. Measurement layer verified.
G1 dose–response	Negative: no structured arm beats MLP at $\geq$2 mission depths with non-decreasing gap.
G2a forced compute	Negative: HRM-v2 k=$1/2/4/8$ expansion and MAE curves flat in every K=8 config.
G2b learned halting	Negative, inverted: pooled Spearman $\rho = -0.407$, permutation $p \approx 5 \times 10^{-4}$, n=675; mean would-halt steps 6.79/7.01/5.30 at K=$2/4/8$.
G3 closure	Neither preregistered branch: global-input arms separate at shallow K (U-Net/GNN vs. MLP ratios $\approx$0.67–0.87; state MAE 0.151/0.171 vs. 0.251 at A/K=0), but no arm converts depth into growing advantage.

Table 10: C11 preregistered gates.

Rung	Factor changed	Δ exp. [95% CI]
C	fresh certifier (oracle rank)	$+11.50$ (0/6 wins)
D	shared-queue certifier (oracle rank)	-6.83 (6/6 wins)
E	exact rollout rank	$+1.33$ (2/6)
F	monotone recalibration	$+1.33$
G	analytic local escape	$+6.83$ (0/6)
I	one-family model, live six suites	$+15.36$ [$+12.02$, $+18.64$]
J	suite-balanced training, static insert	$+16.17$ [$+9.67$, $+22.67$]
K	+ one local Bellman backup	-1.21 [-8.04, $+5.46$]
L	+ scale α=1.50 (48 fresh maps)	-13.04 [-18.67, -7.65]
M	frozen 144-map confirmation	-12.96 [-16.30, -9.74]

Table 11: The C13 falsification ladder. Target repairs (E, F), analytic constructions (G), and distribution repairs (I, J) fail; the integration repair (K) plus scale calibration (L) flip the sign, and the frozen confirmation (M) reproduces it.

Suite	Local	Field HRM	Δ	95% CI	W/T/L
Maze	47.63	84.38	-36.75	[-45.58, -27.83]	24/0/0
Dense maze	95.38	104.50	-9.13	[-15.21, -3.13]	18/1/5
Rooms	98.04	106.17	-8.13	[-14.33, -1.71]	16/0/8
Spiral	120.25	121.33	-1.08	[-5.88, $+4.21$]	14/1/9
Bugtrap	14.96	24.29	-9.33	[-17.79, -2.92]	18/1/5
Large rooms	33.58	46.92	-13.33	[-18.92, -7.50]	19/0/5
Pooled	68.31	81.26	-12.96	[-16.30, -9.74]	109/3/32

Table 12: C13-M per-suite paired expansion deltas versus the complete-map field-HRM comparator. Every suite mean is negative; the spiral interval crosses zero, and the preregistered gate is pooled with a suite-breadth condition (at least four negative suite means), which passes. The local provider also beats every other fixed C7 learned provider in pooled expansions on this cohort.

Cell (HRM)	Δsuccess [CI]	map ratio [CI]
dense LoRA K=1	$+0.467$ [$+0.300$, $+0.633$]	0.686 [0.590, 0.774]
dense LoRA K=16	$+0.433$ [$+0.267$, $+0.600$]	0.786 [0.694, 0.891]
dense full K=1	$+0.227$ [$+0.080$, $+0.380$]	0.789 [0.751, 0.836]
dense full K=16	$+0.460$ [$+0.287$, $+0.633$]	**0.610** [0.549, 0.670]
dense scratch K=1	-0.167 [-0.300, -0.047]	1.035 [1.029, 1.039]
rl. full K=1	-0.167 [-0.340, $+0.013$]	1.293 [1.170, 1.503]
rl. full K=8	$+0.387$ [$+0.227$, $+0.547$]	0.590 [0.498, 0.712]
bug. LoRA K=32	-0.320 [-0.507, -0.133]	1.153 [1.012, 1.523]

Table 13: Map-clustered C9 key cells (binding budgets, additive mode; dense = maze-dense, rl. = rooms-large, bug. = bugtrap). The K=16 full-FT/LoRA CIs are disjoint (the crossover), the rooms-large K=1 catastrophe excludes parity, and the bugtrap LoRA degradation is significant in both metrics.